

The Majority C -Chromatic Number of $K_3 \square K_3 \square K_3$

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Abstract

For the Hamming graph $H(3,3) = K_3 \square K_3 \square K_3$, we prove that the maximum number of colors in a surjective majority C -coloring is four. The upper bound follows from a local observation: an induced subgraph of minimum degree at least three has at least six vertices. The matching lower bound is witnessed by an explicit partition into classes of orders 9, 6, 6, 6. The proof is elementary and does not depend on computation; an exact exhaustive check is described as a reproducibility cross-check.

1 Statement and context

Write $[3] = \{1, 2, 3\}$, and let $H(3,3)$ be the graph on $[3]^3$ in which two vertices are adjacent precisely when they differ in exactly one coordinate. Every vertex has degree six. A *majority C -coloring* is a surjection from the vertex set to a set of colors such that each vertex has at least half of its neighbors in its own color class. We write $\bar{\chi}_{\geq}(G)$ for the maximum number of colors in such a coloring. This coloring model was introduced and studied in [1, 2].

Bujtás, Dettlaff, Furmańczyk, and Laskowska [1, Proposition 6(ii) and Open Problem 3] give bounds for odd-dimensional balanced Hamming graphs and ask for their exact majority C -chromatic numbers. At $n = k = 3$, their bounds specialize to

$$3 \leq \bar{\chi}_{\geq}(H(3,3)) \leq 5.$$

We determine this instance exactly.

Theorem 1. *For the surjective weak-majority convention above,*

$$\bar{\chi}_{\geq}(K_3 \square K_3 \square K_3) = 4.$$

Equivalently, every color class S must satisfy $\delta(H(3,3)[S]) \geq 3$. Thus the theorem amounts to bounding the size of such sets and giving a four-class partition.

2 Proof

We first record two elementary features of the graph.

Lemma 2. *The graph $H(3,3)$ contains no K_4 , and every adjacent pair of vertices has exactly one common neighbor.*

Proof. Fix a vertex v . Its six neighbors split into three pairs according to the coordinate changed from v . The two vertices within any one pair are adjacent, while vertices in different pairs differ in two coordinates and are nonadjacent. A clique containing v therefore has order at most three.

Now let adjacent vertices x, y differ in coordinate i . A common neighbor cannot differ from x outside coordinate i , because it would then differ from y in two coordinates. It must agree with x, y outside i , and its i -th coordinate must be the unique element of $[3] \setminus \{x_i, y_i\}$. This vertex exists and is unique. $\square$

Lemma 3. *If $\emptyset \neq S \subseteq [3]^3$ and $\delta(H(3, 3)[S]) \geq 3$, then $|S| \geq 6$.*

Proof. The assertion is immediate for $|S| \leq 3$. If $|S| = 4$, minimum degree at least three would force $H(3, 3)[S] \cong K_4$, contrary to Lemma 2.

Suppose $|S| = 5$. In the complement of $H(3, 3)[S]$, every vertex has degree at most one, so the missing edges form a matching. If there are zero or one missing edges, four vertices induce a K_4 , again impossible. The remaining case consists of two missing edges, say ab and cd , and a fifth vertex e . Then e is adjacent to a , while both c and d are common neighbors of e and a . This contradicts the uniqueness assertion in Lemma 2. $\square$

Proof of Theorem 1. Let a majority C -coloring use p colors. Surjectivity makes its color classes nonempty, and the majority condition gives induced minimum degree at least three in every class. Lemma 3 shows that every class has at least six vertices. Since the classes partition all 27 vertices, $6p \leq 27$, and hence $p \leq 4$.

For the reverse inequality, define

$$C_0 = \{1\} \times [3] \times [3], \quad C_j = \{2, 3\} \times \{j\} \times [3] \quad (j = 1, 2, 3).$$

These four nonempty sets are pairwise disjoint and cover $[3]^3$. A vertex of C_0 has four neighbors in C_0 , obtained by changing its second or third coordinate. A vertex of C_j has three neighbors in C_j : one obtained by changing its first coordinate within $\{2, 3\}$, and two by changing its third coordinate. Giving the four sets distinct colors is therefore a surjective majority C -coloring. Thus $p \geq 4$, completing the proof. $\square$

3 Exact computational cross-check

The proof above is self-contained. As a separate finite check, the accompanying Python program `verify_resolution.py` constructs the graph from the coordinate definition, examines all

$$\sum_{s=1}^5 \binom{27}{s} = 101,583$$

subsets of order at most five, and finds none of induced minimum degree at least three. It also verifies that the four displayed classes have orders 9, 6, 6, 6, form an exact partition, and have constant induced degrees 4, 3, 3, 3, respectively. All operations use exact integer and set arithmetic.

For methodological context, exact finite certificates and symmetry-reduced checking have also been used for a different coloring problem on a finite ternary coordinate model [3]. That work does not concern majority C -colorings and is not used in the proof here.

Relative to the formulation in [1], Theorem 1 supplies the exact value for the single case $n = k = 3$. It makes no assertion about the remaining odd-dimensional balanced Hamming graphs.

Computational and AI assistance. A generative language model assisted with organizing the proof, checking edge cases, and preparing the manuscript. The exact exhaustive program described above was used as a cross-check. Neither the model output nor the computation is invoked as a substitute for the mathematical proof.

References

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